

A First Course in Stochastic Calculus

Chapter 1

1.6.

$$CDF = P(X^2 \leq x)$$

$$CDF = P(-\sqrt{x} \leq x \leq \sqrt{x}) = \int_{-\sqrt{x}}^{\sqrt{x}} 1/2 dx = \sqrt{x}$$

$$PDF = \frac{d(CDF)}{dy}$$

$$PDF = \frac{1}{2 * \sqrt{x}}$$

1.7.

Integrable means

$$E[|X|] < \infty$$

$$|aX + bY| \leq |aX| + |bY|$$

$$E[|aX + bY|] \leq E[|aX| + |bY|] = E[|aX|] + E[|bY|] < \infty$$

Hence proved.

1.8.

We need to show

$$P(X = a) = F_X(a) - \lim_{x \rightarrow a^-} F_X(x) = F_X(a) - F_X(a^-)$$

$$P(X \leq a - h) = F_X(a - h)$$

$$P(X \leq a + h) = F_X(a + h)$$

$$P(a - h \leq X \leq a + h) = \lim_{h \rightarrow 0} [F_X(a + h) - F_X(a - h)] = F_X(a) - \lim_{h \rightarrow 0} F_X(a - h)$$

$$F_X(a) - \lim_{h \rightarrow 0} F_X(a - h) = F_X(a) - F_X(a^-)$$

1.9.

$$P(Y > t + s | Y > s) = P(Y > t + s) / P(Y > s)$$

$$= \frac{1 - F_X(t + s)}{1 - F_X(s)}$$

$$F(s) = \int_0^s \lambda * e^{-\lambda * y} dy = \int_0^{\lambda * s} e^{-t} dt = [-e^{-t}]_0^{\lambda * s}$$

$$F(s) = 1 - e^{-\lambda * s}$$

$$F(t + s) = 1 - e^{-\lambda * (t+s)}$$

$$\frac{1 - F_X(t + s)}{1 - F_X(s)} = e^{-\lambda * (t+s)} / e^{-\lambda * s} = e^{-\lambda * t} = 1 - F(t) = P(Y > t)$$

1.10.

Integration by parts:

$$\int u dv = uv - \int v du.$$

$$E[Zg(Z)] = \int_{-\infty}^{\infty} z * g(z) * \frac{e^{-z^2/2}}{\sqrt{2\pi}} * dz$$

Applying integration by parts:

$$u = g(z), du = g'(z)$$

$$dv = ze^{-z^2/2} dz$$

$$z^2/2 = t \implies z \cdot dz = dt$$

$$v = \int e^{-t} dt \implies v = -e^{-t}$$

$$= [uv]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} v du$$

$$(1/\sqrt{2\pi})[[-e^{-z^2/2} * g(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} g'(z) * e^{-z^2/2} dz] \implies E[g'(z)]$$

$$\text{B) } E[Z^2] = E[Z \cdot Z] = 1$$

$$E[Z^{2j}] = E[Z \cdot Z^{2j-1}] = E[(2j-1) * Z^{2j-2}]$$

1.11.

$$E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} * \lambda * e^{-\lambda * x} dx \implies \int_{-\infty}^{\infty} \lambda * e^{(-\lambda+t)x} dx$$

We assume, $t < \lambda$

$$\int_{-\infty}^{\infty} \lambda * e^{(-\lambda+t)x} dx = \frac{\lambda}{(t-\lambda)} \int_{-\infty}^{\infty} (\lambda-t) * e^{-(\lambda-t)x} dx \implies \frac{\lambda}{(\lambda-t)}$$

$$E[X] = \frac{\lambda}{(\lambda-t)^2} \implies 1/\lambda$$

$$E[X^2] = \frac{2\lambda}{(\lambda-t)^3} = 2/\lambda^2$$

$$Var(X) = E[X^2] - (E[X])^2 = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$$

$t \geq \lambda$, integral diverges

1.12.

Using Chernoff's bound

$$P(X > a) \leq e^{-\lambda a} E[e^{a\lambda X}]$$

Taking derivative on both side of equation above:

$$E[e^{\lambda X}] \leq e^{\lambda^2/2} \implies e^{-\lambda a} E[e^{a\lambda X}] \leq e^{-\lambda a} * e^{a^2 \lambda^2/2} \implies e^{(a^2/2)(\lambda^2 - 2\lambda/a - 1/a + 1/a)}$$

$$e^{(a^2/2)(\lambda^2 - 2\lambda/a - 1/a^2 + 1/a^2)} \implies e^{(a^2/2)[(\lambda - 1/a)^2 - 1/a^2]}$$

$\lambda = 2/a$ we get the result.

1.13.

$$E[X] = \int_0^{\infty} x f(x) dx$$

$$x = F^{-1}(t) \implies t = F(x) \implies dt = f(x) dx$$

$$\int_0^{\infty} x f(x) dx = \int_0^{\infty} x d(F(x))$$

Integration by parts,

$$u = x, dv = f(x)$$

$$v = 1 - F(x)$$

$$\int_0^{\infty} x d(F(x)) = [x(1 - F(x))]_0^{\infty} - \int_0^{\infty} (1 - F(x)) dx$$

If LHS diverges this is because $1-F(x)$ doesn't decay fast enough. In other case, $x(1-F(x))$ is 0.

$$\int_0^{\infty} (1 - F(x)) dx = \int_0^{\infty} P(X > x) dx$$

For Part B

$$X = X 1_{X \geq 0} + X 1_{X < 0}$$

Taking Expectations on both sides

$$E[X] = E[X 1_{X \geq 0}] + E[X 1_{X < 0}]$$

As per above,

$$E[X1_{X \geq 0}] = \int_0^{\infty} P(X > x) dx$$

$$E[-X1_{X < 0}] = \int_{-\infty}^0 P(X \leq x) dx$$

$$E[X] = \int_0^{\infty} P(X > x) dx - \int_{-\infty}^0 P(X \leq x) dx$$

1.14. a)

$$E[e^{itX}] = E\left[\sum \frac{(itX)^n}{n!}\right]$$

$$E\left[\sum \frac{(itX)^n}{n!}\right] = \left[\sum E[i^n t^n X^n]/n!\right]$$

$$\sum E[i^n t^n X^n]/n! = \sum E[i^{2n} t^{2n} X^{2n}]/2n! + \sum E[i^{2n+1} t^{2n+1} X^{2n+1}]/(2n+1)! \implies \sum E[(-1)^n t^{2n} X^{2n}]/2n! + i \sum E[(-1)^n t^{2n+1} X^{2n+1}]/(2n+1)!$$

$$\sum E[(-1)^n t^{2n} X^{2n}]/2n! + i \sum E[(-1)^n t^{2n+1} X^{2n+1}]/(2n+1)! = E[\cos(tX)] + iE[\sin(tX)]$$

b)

For Gaussian Distribution:

$$tX = mt + \sigma^2 Zt \implies E[e^{itX}] = E[e^{imt+i\sigma^2 Zt}]$$

$$E[e^{imt+i\sigma^2 Zt}] = e^{imt} E[e^{i\sigma^2 Zt}] = e^{imt} \{E[\cos(\sigma^2 Zt)] + iE[\sin(\sigma^2 Zt)]\}$$

$$E[\cos(tZ)]' = E[-Z\sin(tZ)] = -tE[\cos(tZ)] = -tf(t) \implies f'(t) = -tf(t)$$

$$f(t) = e^{-t^2/2}$$

Since $\sin(-x) = -\sin(x)$,

$$E[\sin(\sigma^2 Zt)] = 0$$

$$e^{imt} E[e^{i\sigma^2 Zt}] = e^{imt} \{E[\cos(\sigma^2 Zt)] + iE[\sin(\sigma^2 Zt)]\} = e^{imt} e^{-t^2/2}$$

The proper way to do this is to use contour integrals. Refer to Probability: A Concise course by Rosanov for the full proof.

1.15.

$$\{X = \infty\} = \bigcap_{n \geq 1} \{X > n\} \implies P(X = \infty) = P\left(\bigcap_{n \geq 1} \{X > n\}\right)$$

Using Lemma 1.4

$$P\left(\bigcap_{n \geq 1} \{X > n\}\right) = \lim_{n \rightarrow \infty} P(X > n)$$

Using Markov Inequality

$$\begin{aligned}
P(X > n) &\leq \frac{1}{n}E[X] \\
\lim_{n \rightarrow \infty} P(X > n) &\leq \lim_{n \rightarrow \infty} \frac{1}{n}E[X] \implies \leq 0 \\
P(X > \infty) &= 0 \implies P(X < \infty) = 1
\end{aligned}$$

1.16.

$$\{X > 0\} = \bigcap_{n \geq 1} \{X > \frac{1}{n}\} \implies P(X > 0) = P\left(\bigcap_{n \geq 1} \{X > \frac{1}{n}\}\right)$$

Using Lemma 1.4 and Markov Inequality

$$\begin{aligned}
P(X > 0) &= P\left(\bigcap_{n \geq 1} \{X > \frac{1}{n}\}\right) = \lim_{n \rightarrow \infty} P(X > 1/n) \\
\lim_{n \rightarrow \infty} P(X > 1/n) &= n \cdot E[X] = 0 \implies P(X = 0) = 1
\end{aligned}$$